

Self-Study Assignment 02

Course Name: Client Side Web Development

Course Code: 25EBCC109

Semester: 2

Div: D

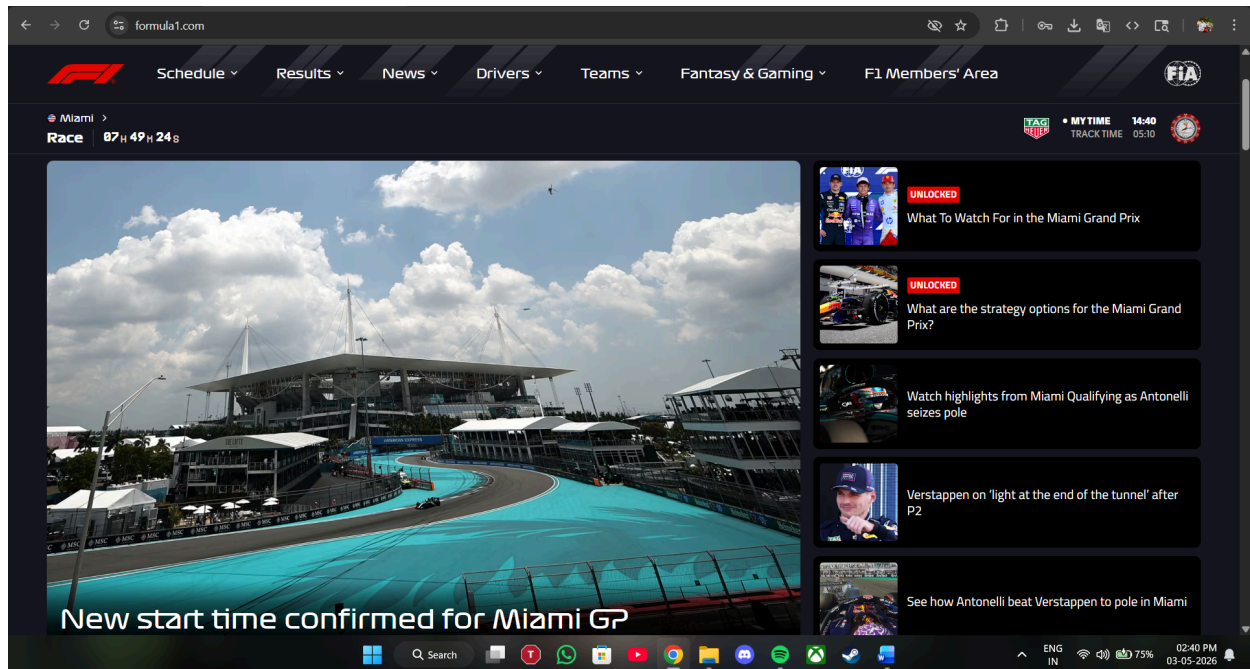
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PART A: Website Information -

The chosen website for this assignment is the official Formula 1 website. It serves as the primary digital hub for all information related to the Formula 1 racing series, including news, schedules, driver standings, and team updates.

Information Type	Details
Name of the Website	Formula 1
URL	https://www.formula1.com/
Protocol	HTTPS
Domain Name	formula1.com
Type of Website	Sports / Entertainment / News
Purpose of the Website	To provide official news, race schedules, live timing, driver/team standings, and exclusive content for Formula 1 fans globally.



PART B: URL Components -

The URL of the website can be broken down into several key components, each serving a specific function in directing the browser to the correct resource.

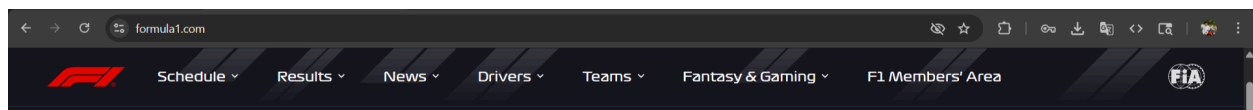
Component	Value	Description
Protocol	<u>https://</u>	Hypertext Transfer Protocol Secure, ensuring encrypted communication.
Subdomain	<u>www.</u>	World Wide Web, a common subdomain indicating a web server.
Domain Name	<u>formula1</u>	The unique identifier for the website.

Domain Extension	<u>.com</u>	Top-Level Domain (TLD) indicating a commercial entity.
Path	<u>/en/results.html/2026/drivers.html</u>	The specific directory and file path leading to the 2026 driver standings page.

PART C: Network Observations –

When a browser requests a webpage, the server responds with HTTP status codes indicating the result of the request.

Common HTTP Status Codes:

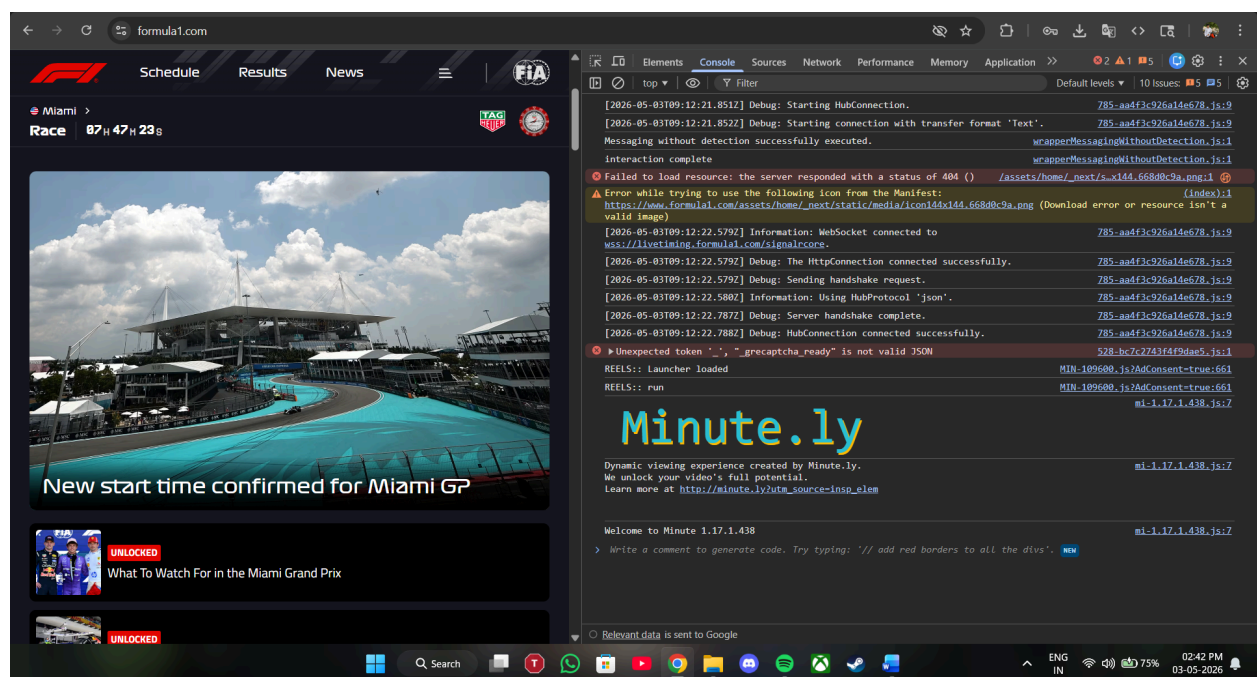


- **200 OK:** The request was successful, and the server has returned the requested data (e.g., loading the homepage).
- **404 Not Found:** The server could not find the requested resource (e.g., navigating to a broken link).
- **500 Internal Server Error:** The server encountered an unexpected condition that prevented it from fulfilling the request.

Network Tab Analysis:

Request / File	Status Code	Meaning
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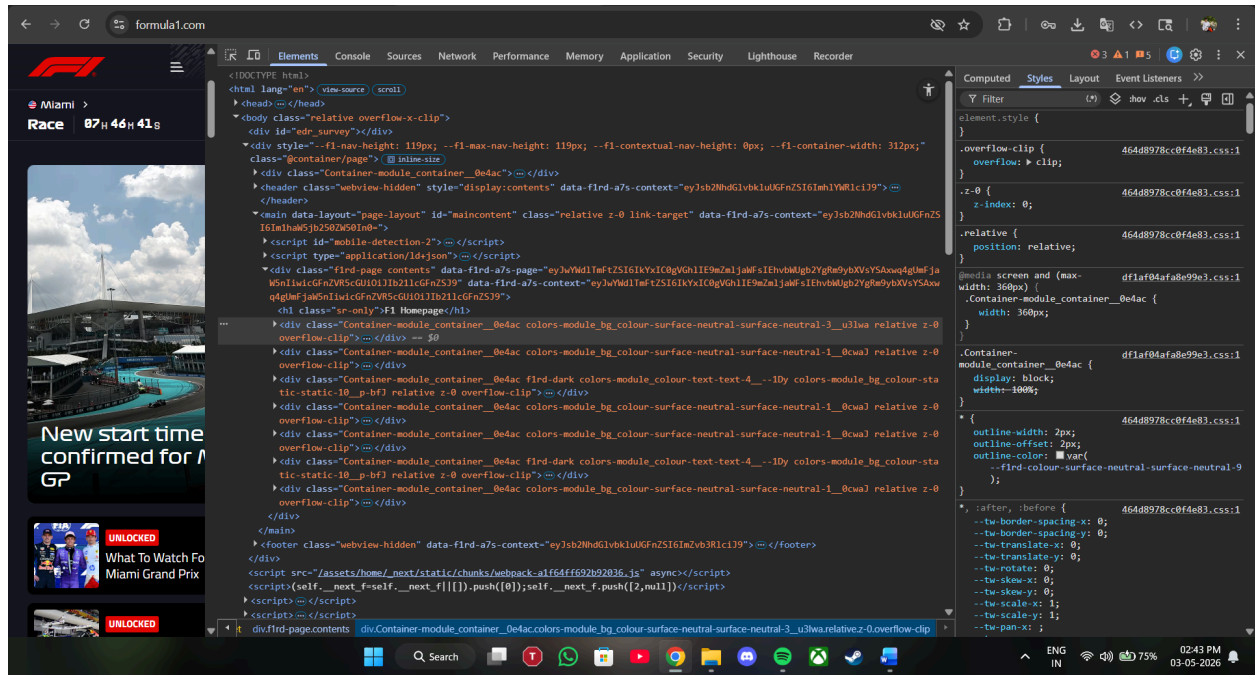
www.formula1.com (Document)	200	The main HTML document loaded successfully.
main.css (Stylesheet)	200	The primary CSS file was fetched successfully.
logo.svg (Image)	200	The site logo image was retrieved successfully.
analytics.js (Script)	200	The tracking script loaded successfully.
non-existent-image.png	404	An image resource could not be found on the server.



PART D: HTML Tags Analysis –

The structure of the Formula 1 website is built using various HTML tags. Below is an analysis of key tags observed on the homepage.

Tag	Where it is used	Purpose
<u><html></u>	Root element	Defines the entire HTML document.
<u><head></u>	Top of document	Contains metadata, title, and links to stylesheets.
<u><body></u>	Main content area	Contains all the visible content of the webpage.
<u><header></u>	Top navigation	Contains the main logo, navigation menu, and sign-in links.
<u><nav></u>	Menu bar	Defines the navigation links (Schedule, Results, News, etc.).
<u><h1></u>	Main headings	Used for primary titles like "F1 - The Official Home of Formula 1® Racing".
<u><a></u>	Hyperlinks	Used for all clickable links, such as driver profiles and news articles.
<u></u>	Images	Used to display article thumbnails, logos, and driver portraits.
<u><div></u>	Layout containers	Used extensively to group elements and apply CSS styling.
<u><footer></u>	Bottom of page	Contains copyright info, privacy policy links, and social media icons.



PART E: Form Elements -

Forms are crucial for user interaction, such as searching, logging in, and registering. The Formula 1 website utilizes several form elements.

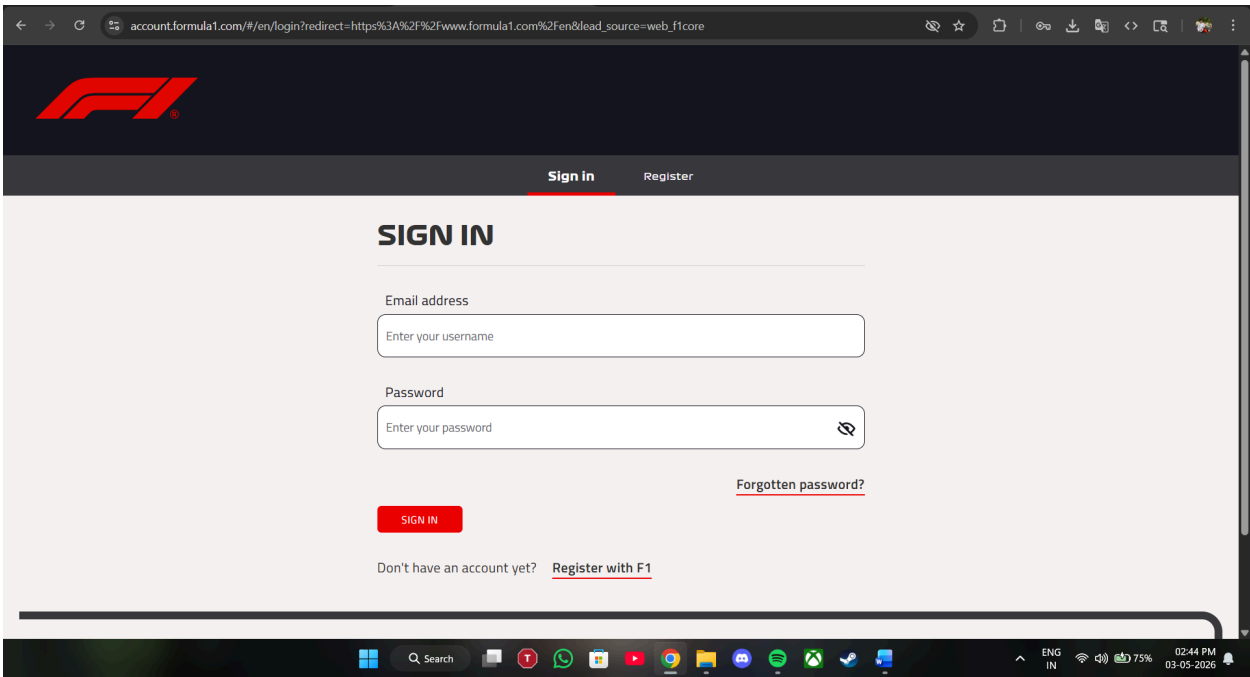
Form Element	Tag Used	Purpose
Text Input	<u><code><input type="text"></code></u>	Used for entering username/email in the login and registration forms.
Password Input	<u><code><input type="password"></code></u>	Used for securely entering passwords, masking the characters.
Dropdown Select	<u><code><select></code></u>	Used in the registration form for selecting Title (Mr, Mrs) and Country.

Submit Button	<code><button type="submit"></code>	Used to submit the login or registration data to the server.
Search Input	<code><input type="search"></code>	Used in the search bar to query the site for specific content.

SEARCH BOX:



SIGN IN FORM:



SIGNUP FORM:

PART F: CSS Properties –

Cascading Style Sheets (CSS) dictate the visual presentation of the HTML elements.

Web Component	CSS Property	Value	Description
Main Header	<u>background-color</u>	<u>#15151e</u>	Sets the dark background color for the top navigation bar.
Primary Buttons	<u>color</u>	<u>#ffffff</u>	Sets the text color of buttons to white.
Primary Buttons	<u>background-color</u>	<u>#e10600</u>	Sets the background color to the signature F1 red.
Navigation Links	<u>font-family</u>	<u>\'Formula1\\', sans-serif</u>	Applies the custom F1 font to the navigation text.

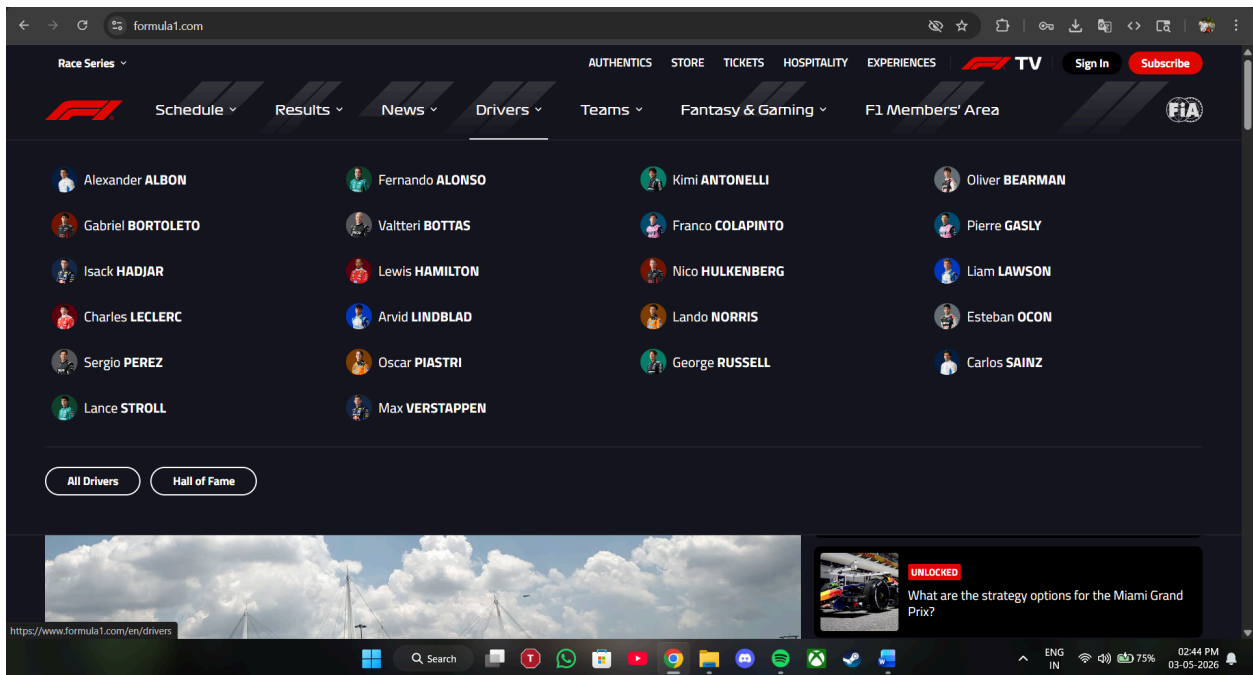
Article Images	<u>border-radius</u>	<u>8px</u>	Rounds the corners of the news article thumbnail images.
Body Text	<u>font-size</u>	<u>16px</u>	Sets the standard readable text size for paragraphs.
Main Container	<u>max-width</u>	<u>1200px</u>	Restricts the maximum width of the content area for readability on large screens.

PART G: Interactive Actions –

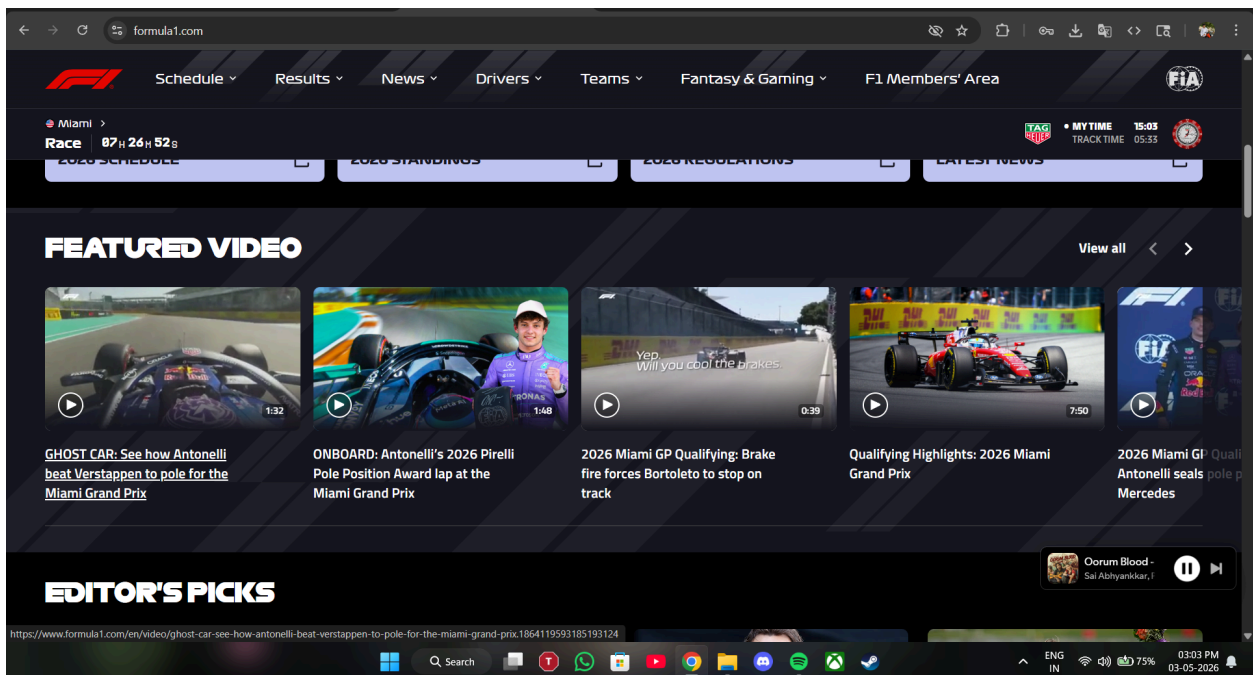
The website includes various interactive elements powered by JavaScript to enhance user experience.

Feature	Action	Event Type	Output
Navigation Menu	Hover over "Drivers"	<u>onmouseover</u>	A dropdown menu appears showing a list of current F1 drivers.
Search Icon	Click the magnifying glass	<u>onclick</u>	The search input field expands or a search modal opens.
Video Player	Click "Play" on a highlight	<u>onclick</u>	The video begins playing within the embedded player.
Cookie Banner	Click "Accept All"	<u>onclick</u>	The cookie consent banner disappears, and preferences are saved.
Image Carousel	Click "Next" arrow	<u>onclick</u>	The carousel slides to display the next set of images or articles.

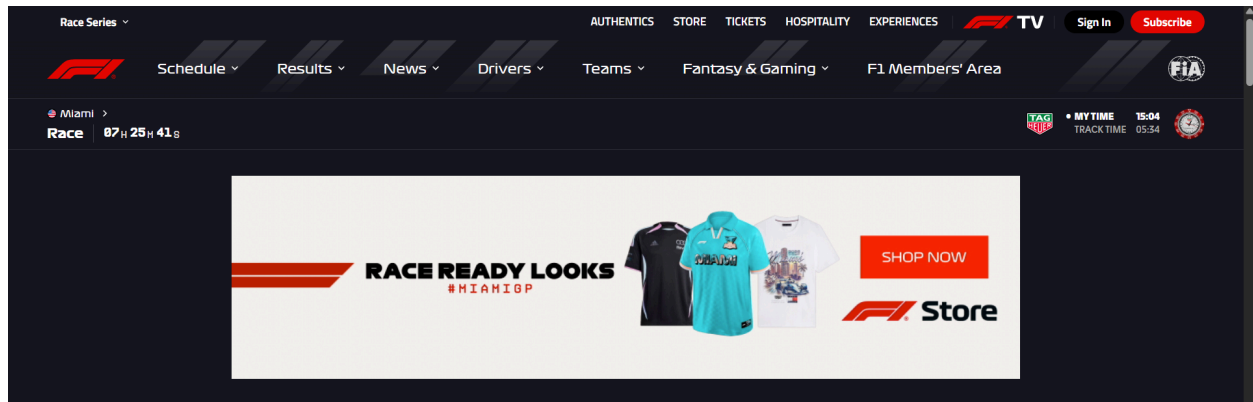
DROPDOWN MENU:



VIDEO PLAY:



SHOPPING SUGGESTIONS:



PART H: Security Questions –

1. Does the website use HTTPS? How do you know?

Yes, the website uses HTTPS. This is evident from the URL, which begins with <https://>, and the padlock icon displayed in the browser's address bar, indicating a secure, encrypted connection.

2. Does the website use cookies? If yes, what is their purpose?

Yes, the website uses cookies. Upon first visit, a cookie consent banner appears. The cookies are used for essential site functionality, storing user preferences, personalizing content and advertisements, and analyzing site traffic to improve performance.

3. Does the website have a login feature? If yes, what information is required?

Yes, the website has a login feature for the "F1 Members' Area" and "F1 TV". It requires an Email address (or username) and a Password.

4. Does the website use CAPTCHA or OTP for verification?

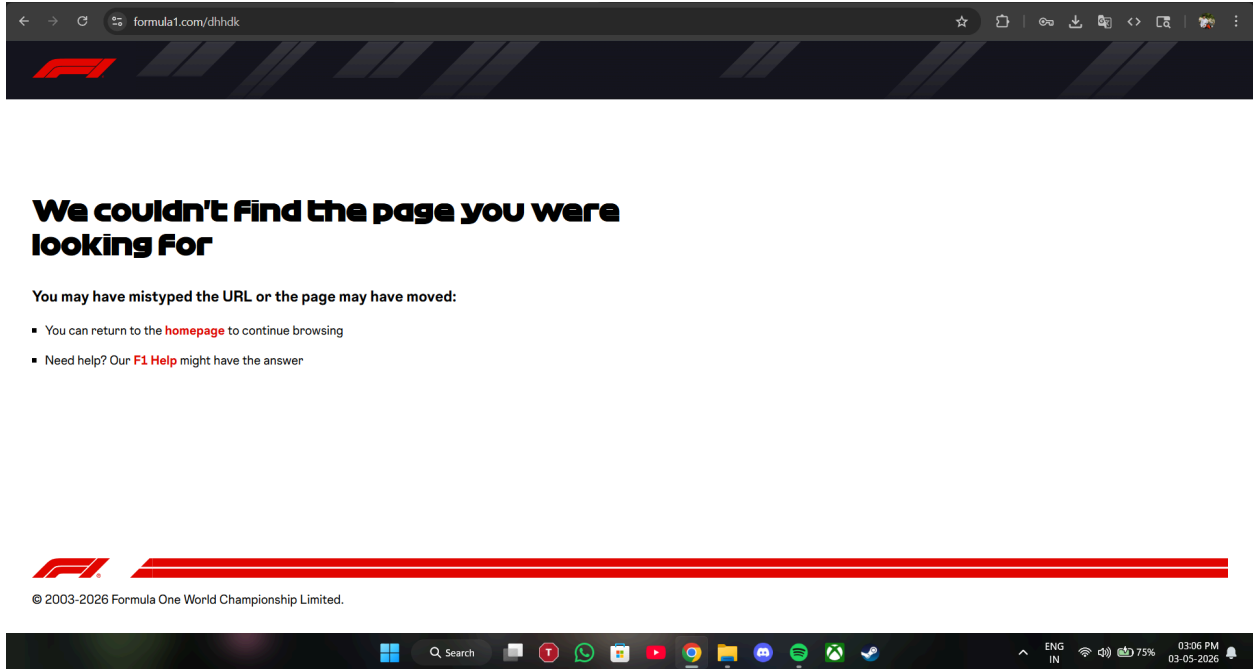
During the standard login process observed, a visible CAPTCHA was not immediately present on the initial form, but modern sites often employ invisible reCAPTCHA or trigger verification steps (like OTP via email) if suspicious activity or a new device is detected.

5. Identify two security features implemented on the website.

- Data Encryption (SSL/TLS): The use of HTTPS ensures that all data transmitted between the user's browser and the server (such as login credentials) is encrypted and protected from interception.
- Password Masking: The password input field uses `<input type="password">`, which masks the characters entered by the user, preventing shoulder surfing.

PART I: 404 Page Analysis -

When navigating to a non-existent URL on the Formula 1 website (e.g., <https://www.formula1.com/dhhdhk>), the server returns a custom 404 error page.



404 Page Details :

- **Message Displayed:** "We couldn't find the page you were looking for. You may have mistyped the URL or the page may have moved."
- **Status Code:** The network tab confirms a 404 Not Found status code is returned.
- **Redirect Behavior:** The site does not automatically redirect to the homepage. Instead, it provides helpful links, such as "return to the homepage" and a link to "F1 Help", allowing the user to choose their next action.